Evaluating the Association Between Polycyclic Aromatic Hydrocarbon Exposure, Epigenetic Age Acceleration, and Lung Cancer Risk

A Systematic Review and Evidence Map

CSR-x Biomedical and Health Sciences Cohort #2

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Background

Polycyclic Aromatic Hydrocarbons (PAHs) are a class of organic compounds composed of two or more fused aromatic rings [24]. Humans are exposed to PAHs through occupational sources of coke-oven work [9,10,20], coal mining [22], and chimney sweeping [25,27,31], or through everyday sources like vehicle emissions [6,12], tobacco smoke [5,16], and charred meats [2]. PAHs can be metabolized into reactive intermediates that bind to DNA and cause mutations [1,17,24] or indirectly trigger oxidative stress and inflammation [2,22,28]. DNA methylation (DNAm) is an epigenetic modification in which a methyl group is primarily attached to the cytosine residue within a cytosine-guanine (CpG) site [18,27]. DNAm patterns can be used to calculate epigenetic age, an estimate of biological aging relative to chronological age [7,9,16]. Few studies have examined the full pathway linking PAH exposure, epigenetic age acceleration, and DNAm alterations with lung carcinogenesis [7,23,29,32].

Research Question:

How is PAH exposure associated with epigenetic age acceleration and lung cancer risk?

Methodology

1. Conducted a systematic literature review of PAH exposure, DNAm, epigenetic aging, oxidative stress, DNA damage, and lung cancer with evidence mapping, cross-study synthesis, and heterogeneity analysis.
2. Searched PubMed, ScienceDirect, and Google Scholar using keywords including PAHs, benzo[a]pyrene, DNA methylation, epigenetic aging, oxidative stress, DNA damage, and lung cancer.
3. Screened studies for PAH exposure and relevant DNAm, epigenetic-aging, mechanistic, or lung cancer outcomes; excluded reviews, duplicates, irrelevant studies, and inaccessible full texts.
4. Extracted study design, population/model, PAH exposure, biomarkers, tissue, methylation methods, biological effects, statistical significance, confounders, strengths, limitations, and evidence type into a standardized evidence matrix.
5. Synthesized 33 peer-reviewed studies with full-text availability.

Results

Across the included studies, PAH exposure was associated with altered DNAm, accelerated epigenetic age, oxidative and genomic damage, and increased lung cancer risk. Evidence was strongest for the direct PAH–lung-cancer association, oxidative stress, and DNA-damage mechanisms, while the complete pathway from PAH exposure through epigenetic age acceleration to lung cancer remained less directly established.

Key Sources/ Acknowledgements

PRISMA Diagram

Study Characteristics

STUDIES	STUDY DESIGNS	PAH EXPOSURE SOURCES	SAMPLE TYPE
33 total studies: human studies were prioritized + mechanistic evidence (animal/cell studies)	Case-control, cross-sectional, longitudinal, and repeated measures + some randomized controlled trial or in vitro designs (focused on PAH-EAA-lung cancer mechanisms)	Occupational (coke-oven/chimney), household (smoky/smokeless coal), tobacco smoke, and air pollution	Blood, urine, lung tissue/cells, and buccal cells

Heterogeneity: Studies varied by population/model, exposure source and measure, DNAm clock, sample type, biological mechanisms/biomarkers, and outcome.

Evidence Maps

Figure 1: PAH Exposure, DNA-Methylation Alterations, and Lung-related Outcomes.

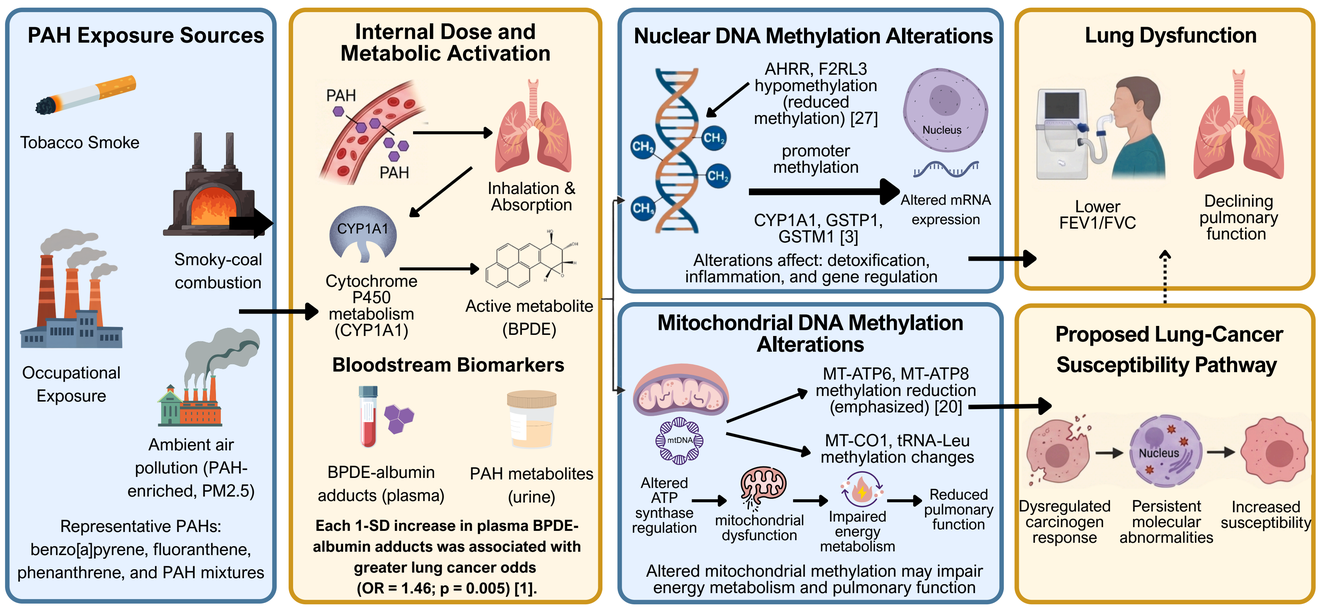


Figure 2: PAH exposure, Epigenetic Age Acceleration, and Lung Cancer Risk.

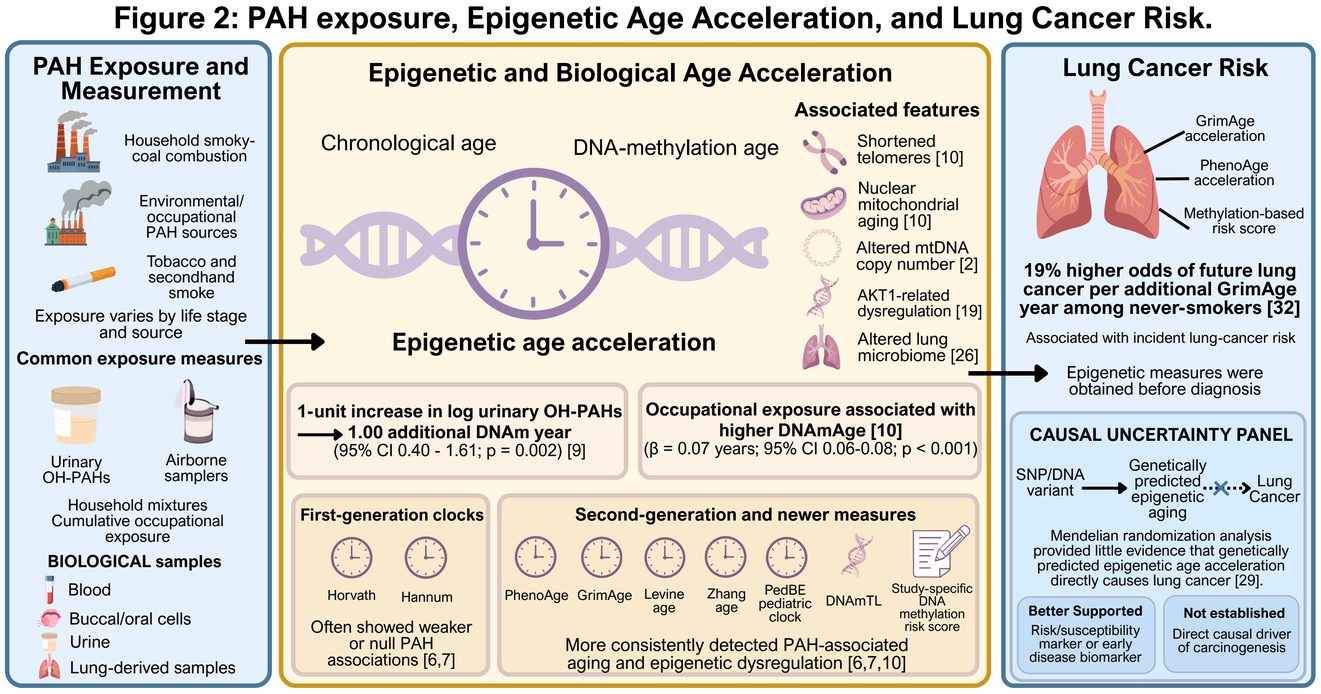
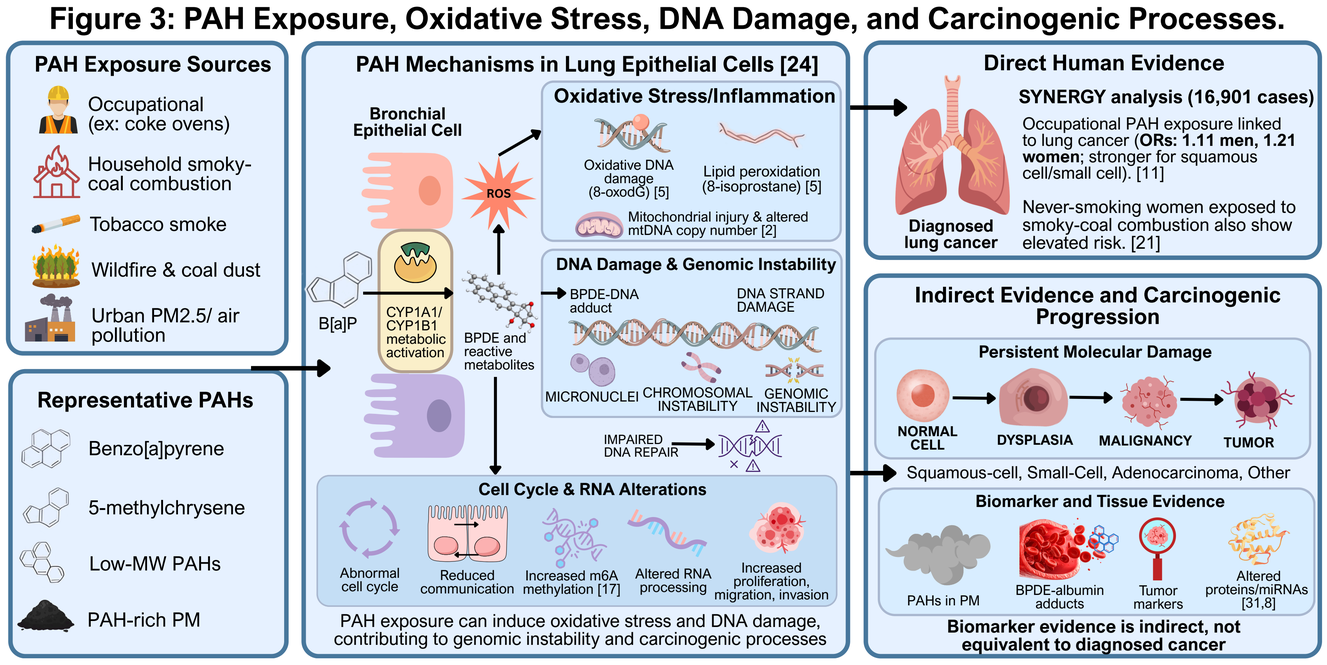


Figure 3: PAH Exposure, Oxidative Stress, DNA Damage, and Carcinogenic Processes.



Discussion

- Our review suggests a biologically plausible model of association between PAH exposure and lung cancer risk.
- Direct human evidence was statistically strongest for PAH exposure → lung cancer risk, while mechanistic studies strongly supported molecular progression through intermediate processes such as oxidative stress and genomic instability.
- PAH exposure → epigenetic age acceleration showed moderate to strong evidence, especially with second-generation DNAm clocks (GrimAge and PhenoAge).
- Epigenetic aging → lung cancer risk was supported as an association and a potential predictive marker, but not a proven cause; Mendelian randomization findings weakened causal certainty.
- PAH exposure links to oxidative stress, genomic instability (DNA damage, faulty repair mechanisms, cell-cycle disruption), DNAm, and epigenetic age acceleration, leading to lung cancer risk, progression, and other lung dysfunction outcomes.
- Varied demographics (location, exposure source, smoking status, age, sex) and measurements (urinary metabolites, BPDE-adduct measurements, gene-specific methylation, epigenetic clocks, lung cancer outcomes) made studies less comparable.
- No single study has shown the full pathway; it was reconstructed using complementary evidence.

Limitations

- Most studies were cross-sectional or case-control, limiting conclusions about long-term causality.
- Many studies relied on urinary metabolites, reflecting recent rather than chronic exposure.
- Short-term measures do not align well with the long developmental timeline of lung carcinogenesis.
- Heterogeneity in exposure source and measurement, methylation clock, outcome definition, population, confounder adjustment, and experimental model reduces direct comparability.
- Occupational cohorts were overrepresented, while some populations were understudied, limiting generalizability.
- Several studies measured carcinogen biomarkers or cellular effects rather than lung cancer incidence, weakening direct disease-specific conclusions.
- PAHs measured were often mixtures, making it difficult to isolate the effects of individual compounds.
- More research is needed to: address underrepresented populations, test the association between epigenetic aging and lung cancer, and address the whole pathway in one cohort.

Conclusion

Environmental PAH exposure is associated with epigenetic age acceleration, while mechanistic evidence supports DNA adduct formation, oxidative stress, inflammation, and abnormal DNAm as intermediate processes. These biological changes are correlated with weakened DNA repair, disrupted tumor-suppressor pathways, and increased genomic instability, creating conditions that support lung carcinogenesis. Human studies have associated PAH exposure with epigenetic age acceleration, while prospective studies have associated accelerated epigenetic aging, particularly GrimAge acceleration, with greater lung cancer risk. Although direct causality has not yet been proven, epigenetic aging is a promising risk biomarker and potential contributor to PAH-associated lung cancer risk.